Q1: Machine learning is a branch of Artificial intelligence where systems learn from data and improve automatically without being explicitly programmed. It includes supervised learning, unsupervised learning and reinforcement learning.

Q2: Supervised learning uses labeled training data to learn a relationship between input and output. It is used for classification and regression problems.

Q3: Unsupervised learning deals with unlabeled data and finds hidden patterns. Clustering and association techniques are common methods.

Q4: Natural Language processing is a field of AI that allows machines to understand human language. Applications include chatbots, machine translation, sentiment analysis and speech recognition.

Q5: Data preprocessing is the process of cleaning and preparing data before training. It includes handling missing values, normalization, encoding and noise removal.